

GRAYWATER ON THE GRID

On October 24, 2006, the EcoHouse, a demonstration green living home in Berkeley, California, gained approval for its constructed wetland and graywater hybrid system. The graywater system became the first to pass code in the state of California, as well as the first permitted, residential graywater system approved by the city of Berkeley. The graywater system also includes the state's first constructed wetland approved for residential use.

Water Wise

Operated through Berkeley's Ecology Center, EcoHouse is a home for three full-time residents that also serves as a demonstration facility for green living. EcoHouse includes features such as solar panels, a tankless water heater, and an organic garden. The center estimates that EcoHouse will save 18,000 to 27,000 gallons of fresh drinking water annually by diverting water from showers, washing machine, and bathroom sinks. The water will be diverted to irrigate a backyard garden (see Figure). Currently, the house uses approximately 75 gallons of water a day. The East Bay Municipal Utility District will track the water usage at EcoHouse, including water saved from the graywater system, by monitoring a backyard meter.

Graywater is defined as water that is sourced from bath, dish, and laundry water. This excludes toilet water—which is full of nitrogen and contains much higher levels of pollutants and bacteria. While the phosphorous, potassium, and nitrogen content of graywater makes it a source of pollution for lakes, rivers,

and ground water, it makes it an excellent source of nutrients for vegetation. Under proper maintenance, this makes graywater valuable when it is used to irrigate gardens and lawns.

Graywater systems have been used in one form or another for centuries, but careful monitoring of the systems, as well as technological



The graywater system will save the EcoHouse 18,000–27,000 gallons of water annually.

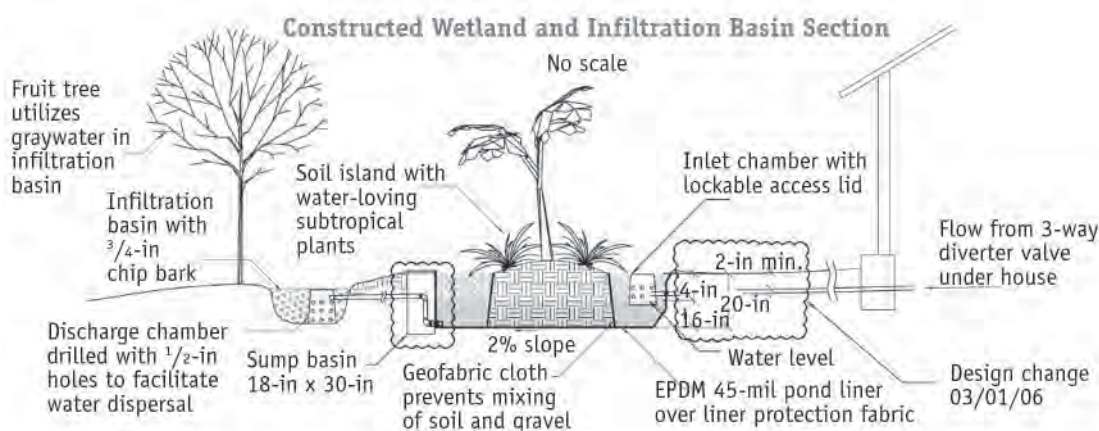


Figure. Water from the EcoHouse's showers, washing machine, and bathroom sinks is diverted to a constructed wetland and infiltration basin.

advances, are recent developments. Unsurprisingly, getting a graywater system approved with a constructed wetland is no small task.

In particular, Berkeley officials were worried that mosquitoes, which can carry such deadly viruses as West Nile, would lay eggs in the graywater system. However, through the use of a design approved by EPA, EcoHouse was able to move an approved design by city officials. Designed by John Russell and the DIG Cooperative, an Oakland design-build group, the system and accompanying wetland was built by 50 EcoHouse volunteers at a cost of approximately \$5,000.

The pondlike wetland is 10 feet in diameter. It is filled with gravel and planted with iris, cattails, and papyrus. After water passes through the wetland, it irrigates four organic garden boxes and several fruit trees.

Eco House

Berkeley EcoHouse, founded in April 1999, is a nonprofit, community-based educational organization created by a group of diverse, talented, and inspired individuals with a common passion for restoring our ecological systems and building healthy, socially just, and stable communities.

The EcoHouse structure is a modest home in an urban neighborhood that, through its renovation, embodies simple environmental approaches to construction. EcoHouse incorporates ecologically friendly materials and methods that reduce resource use, come from renewable resources, are manufactured in ways that have a minimal impact on the environment, contain no toxic chemicals, do not off-gas harmful compounds, and can be reused or reclaimed at the end of their useful life. It contains such environmentally preferable products as natural linoleum floors, wood floors salvaged from a demolished house, salvaged cabinets, sustainably harvested bamboo for the kitchen countertops, and wood from sustainable forests.

Renovations on the home included installing cellulose insulation, replacing lighting fixtures with low-energy CFLs, installing high-efficiency appliances, and replacing the standard water heater with a flash, or instantaneous, water heater that heats water only as it is needed. A PV installation provides all of the home's electricity.

EcoHouse teaches a solar energy curriculum to over 1,500 students in Berkeley schools. Students learn math skills to calculate seasonal sun angles; they design and build models of solar-heated homes; and they learn to use PV solar panels to power music systems, toys, lights, and irrigation pumps. EcoHouse also sponsors Youth Energy Services (YES), which trains high school students to perform energy audits in the homes of seniors and the disabled and at homeless shelters, and then retrofit those homes for energy conservation.

With educational, environmental, and energy efficiency measures in place, the graywater system was a natural next step for EcoHouse. Resident Babak Jacinto Tondre, who is the home's graywater coordinator, says that while some people believe that conserving energy will be the most difficult challenge we face in the twenty-first century, conserving water will be an integral part of that challenge. By practicing tough conservation measures and innovative approaches to water use—such as EcoHouse's graywater system—we stand to come a little closer to becoming wise water users in our homes. 

—Elka Karl

*Elka Karl is an associate editor at **Home Energy** magazine.*

For more information:

To learn more about EcoHouse, go to
www.ecohouse.org.